Project Proposal

1. Project Name

Supply Chain Performance Analysis: Shipment Delays, Cost Efficiency, Inventory Availability, and the Impact of Shortages on Patient Cases

2. Project Description and Implementation

2.1 Objective

Measure what the data actually supports: the delivery cycle and cost of the historical shipments in the shipments table (in), followed by the analysis of purchase orders, inventory and patient cases in the hospital layer, and convert the results into decisions supported by numbers that can be recalculated. The project pursues three objectives: (1) describe the shipment delivery cycle and cost: schedule status, lead time, customs clearance, shelf life at delivery, vendor value and freight cost; (2) measure supplier performance, stockout, unmet demand and wastage in the hospital layer (purchase orders and daily inventory); (3) describe patient case outcomes by treatment status in the hospital layer and their descriptive link to item stockouts, without causal claims.

2.2 Description

A five-page Power BI analysis of supply chain performance, delivered in the Data Analytics track. The analysis answers 21 analytical questions (Q01 to Q21) with 38 visuals; each question is tied to specific columns, a written KPI definition and a chart that the source data can support. The source is one workbook (Data_Final_project_IR.xlsx) with two separate data families:

- **Shipments (sheet in):** 46 data rows, 37 columns, 46 unique id values, 40 distinct po / so # values, 7 countries, 11 vendors and 3 product groups; the scheduled, delivered and recorded delivery dates span 7 November 2007 to 18 August 2015 (the earliest PO sent to vendor date is 2 October 2007).
- **Hospital layer:** items (30 items: 14 Critical, 16 Essential), suppliers (9), hospitals (4), inventory_daily (10,793 rows), purchase_orders (2,187 rows) and patient_cases (5,665 rows); inventory dates run from 1 January to 31 March 2026 according to the data dictionary.
- **Reference sheets:** data_dictionary (59 definitions), schema_relationships (9 relationships), kpi_definitions (11 definitions) and issues_answer_key (107 documented issues).

2.3 Boundaries applied throughout the project

- No figure is written from memory. Every reference number is calculated directly from the source file and recalculated after cleaning, with the effect of each change documented.
- All 46 shipments in sheet in were delivered on or before the scheduled date (44 on schedule, 2 earlier, 0 later) and shipment mode is Air in 45 rows (1 blank). The shipment analysis is therefore built on schedule status, lead time, customs clearance, shelf life, vendor value and freight cost, not on delay rankings or mode comparisons.
- Shipments and hospital tables are separate families. There is no direct relationship or causal inference between them, and no historical shipment is linked to a patient or a hospital.
- The hospital layer is documented in the data file as fictional/synthetic. It is presented as an illustrative analytical case, not as medical evidence that a supplier delay caused a death or a complication.

- Blank values remain NULL and are never converted to zero. Outliers are verified, flagged and explained; they are not deleted because they are large.
- Small groups are not ranked: countries with fewer than 5 shipments and vendors with fewer than 3 shipments are grouped as Other, and every visual shows its sample size.
- Data quality rules follow the approved Data Quality Rules: fully identical duplicates are removed only after logging; slash dates are parsed explicitly as dd/mm/yyyy; rows outside 1 January to 31 March 2026 are flagged Out_of_Period and excluded only from time-sensitive KPIs; S09 (regional distribution centre) is excluded from commercial supplier rankings; shortage-type treatment status is the exact match of "Delayed - stock unavailable", "Substitute used" and "Not given".

2.4 Tools and Technologies

Excel for manual inspection and documentation; SQL for cleaning, validation and repeatable calculations; Python 3 (pandas, NumPy, matplotlib, SciPy and statsmodels) for recalculating the main KPIs, drawing descriptive charts and running the statistical analysis; Power BI Desktop (Power Query, Model view, DAX, Report view) for the model and visuals; GitHub for version tracking (one repository with the folders sql, python, powerbi, data and docs; a main branch plus one branch per member; pull request before merge); Microsoft Word/PowerPoint or PDF for documentation and presentation.

2.5 Dashboard Design and Data Flow

- **Dashboard:** five pages - Executive Overview (5 visuals); Delivery & Cycle Time (7); Suppliers & Cost (8); Purchase Orders & Inventory (9); Case Impact & Decision Quality (9) - covering questions Q01 to Q21 with 38 visuals (KPI cards, column, bar, Pareto, histogram, scatter, box plot, heatmap, line, donut and tables).
- **Data flow:** source files, Raw layer, Working layer, SQL Staging, Power BI models (shipments and hospitals kept separate), dashboard pages.
- **Execution principle:** for every result - source and specific column, documented cleaning rule, written KPI definition, SQL/Python calculation, comparison with Power BI, and an explanation of the limits of the conclusion.

2.6 Python Scope and Statistical Analysis

One notebook loads the cleaned dataset, recomputes the headline KPIs of each family, draws basic histograms with an outlier check (IQR rule), runs the statistical analysis below and compares the results with the SQL and Power BI figures. The libraries are pandas and NumPy (data handling), SciPy (scipy.stats, statistical tests), statsmodels (confidence intervals, multiple-comparison correction, stratified tables) and matplotlib (charts). The notebook runs from top to bottom with no manual step, uses a fixed random seed (42) for every resampling and writes a results table (stats_results.csv) with the columns step, question, n_valid, n_excluded, method, statistic, p_value, p_holm, effect_size, ci_low and ci_high.

Rules for every step: the valid-row definitions of the approved Data Quality Rules are used as denominators (PO Delay and PO On-Time 1,962 rows; Stockout Rate 8,926 rows; Lead Time 45 rows; Freight per Kg 38 rows); each test drops only rows that are NULL in its own columns and reports the excluded count; every p-value is shown with its sample size, an effect size and a 95% confidence interval; comparisons within one family are corrected with the Holm method; the significance level is 0.05 and the minimum group size for hospital-layer groups is 30 rows or cases (both documented assumptions, confirmed in the Question Map); results are descriptive associations and make no causal claim.

Step	Questions	Analysis and data used	Method and library function
S1	Q04, Q05, Q07, Q11	Descriptive statistics of Delay Days (46 rows: 0 late, 44 on schedule, 2 early), Lead Time Days (45), Shelf Life at Delivery and Freight per Kg (38). No inferential test on Delay Days because no shipment is late.	pandas describe and quantile (median, IQR, P90); scipy.stats.skew; statsmodels proportion_confint (Wilson).
S2	Q05	Lead time by country, for countries with at least 5 valid lead times: Nigeria 17, Haiti 7, Cote d'Ivoire 6, Congo DRC 6, Guyana 5; the rest grouped as Other.	scipy.stats.kruskal with epsilon-squared; pairwise scipy.stats.mannwhitneyu with Holm correction.
S3	Q06	Customs clearance days against Lead Time Days on the 44 rows where both are valid ("3 days" and "~6" converted and flagged); median share of lead time spent in customs.	scipy.stats.spearmanr; scipy.stats.bootstrap (10,000 resamples) for the interval; sensitivity run without approximate values.
S4	Q07	Shelf life at delivery (valid expiry dates only) and receiving inspection results: share expired on arrival and Inspection Acceptance Rate.	pandas; statsmodels proportion_confint (Wilson); scipy.stats.skew.
S5	Q08	Vendor concentration of shipment value (Pareto), vendors below 3 shipments grouped as Other.	pandas groupby and cumulative share; NumPy Herfindahl-Hirschman Index.
S6	Q10, Q11	Freight against weight on the 38 rows with both values numeric; Freight/Value on rows with numeric freight; outliers on Freight per Kg flagged, not deleted.	scipy.stats.spearmanr; scipy.stats.linregress on log values; IQR rule (1.5 x IQR).
S7	Q12	Actual on-time rate of each supplier S01 to S08 (1,961 of the 1,962 valid received POs; S09 excluded) against suppliers.historical_on_time_rate; PO Fill Ratio.	scipy.stats.binomtest per supplier; statsmodels proportion_confint (Wilson); Holm correction.
S8	Q13	PO Delay Days distribution and differences between suppliers; status mix reported as counts, Rejected orders reported separately.	scipy.stats.kruskal with epsilon-squared; pairwise scipy.stats.mannwhitneyu with Holm correction.
S9	Q14	Actual order-to-receipt lead time against items.baseline_lead_time_days by item category, received POs with valid dates.	scipy.stats.wilcoxon on paired differences; rank-biserial effect size; Holm correction.
S10	Q15	Stockout Rate (unmet_demand_qty > 0) on 8,926 valid hospital-item-days by hospital and criticality, and week by week; recomputed flag compared with the source stockout_flag.	scipy.stats.chi2_contingency with Cramer's V; Wilson intervals; scipy.stats.kendalltau for the weekly trend; pandas.crosstab.
S11	Q16	Estimated value of unmet demand (top 10 items) and wastage by category and cold-chain flag, valued at items.unit_cost_egp; descriptive only because only 3 of 30 items are cold-chain items.	pandas groupby and cumulative share.
S12	Q17	Treatment delay hours by treatment status; values of 999 hours (14 rows) and negative values (14 rows) are flagged, not deleted; "Not given" has no numeric delay values.	scipy.stats.kruskal; pairwise scipy.stats.mannwhitneyu against "Given on time" with Holm correction.

Step	Questions	Analysis and data used	Method and library function
S13	Q18	Death share, ICU admission, complication and length of stay by treatment status within each severity level; shortage-type against "Given on time"; Shortage Mortality = Died / (Died + Recovered).	scipy.stats.fisher_exact per severity; statsmodels StratifiedTable (Mantel-Haenszel odds ratio, Breslow-Day test); scipy.stats.mannwhitneyu.
S14	Q19	Item Stockout Rate against the item share of "Delayed - stock unavailable" cases, valid items I001 to I030 with enough cases (19 valid items in the cases table).	scipy.stats.spearmanr; scipy.stats.bootstrap; Wilson intervals.
S15	Q20	Shortage-type share by diagnosis (documented mapping to 13 groups) and by hospital H-A to H-D, above the minimum group size.	scipy.stats.chi2_contingency with Cramer's V; adjusted standardized residuals; Holm correction.
S16	Q21, Q03	Stability of the key results after cleaning: each test run on all valid rows and after excluding flagged rows.	Same functions as the original steps; change of estimate, direction and significance.
S17	All	Reporting and reproducibility: results table, charts and library versions saved in the python folder of the repository.	pandas to_csv; matplotlib savefig; numpy default_rng(42); pip freeze.

2.7 Weekly Implementation Plan

Week 1 (11-15 October 2026): Scope, Definitions, Raw/Working Layers and Planning Documents

- **Hossam:** KPI Specification and KPIs Document; GitHub Repository and Branching Strategy; Project Plan and Task Assignment; Project Proposal.
- **Yahia:** Raw and Working Data Layers with Data Profiling.
- **Ola:** Stakeholder Analysis and User Stories & Use Cases; Literature Review File (Lecturer Input).
- **Aya:** Data Dictionary; Functional & Non-functional Requirements; Risk Assessment & Mitigation Plan.

Deliverables: KPI Specification + KPIs Document; Repository + Branching Strategy; Raw/Working Data Layers + Profiling Report; Project Plan + Task Assignment & Roles; Project Proposal; Stakeholder Analysis + User Stories & Use Cases; Literature Review; Validated Data Dictionary; Functional & Non-functional Requirements; Risk Assessment & Mitigation Plan.

Exit gate: Planning, literature review and requirements documents are ready for submission on 16 October 2026.

Week 2 (18-22 October 2026): Data Cleaning, Integrity Checks and Quality Rules

- **Hossam:** SQL Staging, Duplicate and Key Integrity Checks.
- **Yahia:** Shipments Table Cleaning; Hospital Tables Cleaning.
- **Ola:** Data Quality Rules and Mapping Sign-off.
- **Aya:** Before/After Reconciliation.

Deliverables: SQL Staging Layer + Integrity Report; Clean Shipments Table + Cleaning Log; Clean Hospital Tables + Cleaning Log; Data Quality Rules + Categorical Mapping Sign-off; Before/After Reconciliation Report.

Exit gate: Cleaning logs and documented mappings are delivered; before/after differences are explained.

Week 3 (25-29 October 2026): Derived Fields, Cleaned Dataset, Python Notebook and Dashboard Design

- **Hossam:** Python Notebook: KPI Recalculation, Basic Charts and Statistical Analysis (steps S0 to S17 of section 2.6).
- **Yahia:** Derived Fields and Cleaned Dataset.
- **Ola:** Question-to-KPI-to-Chart Map.
- **Aya:** Wireframes & UI/UX Guidelines.

Deliverables: Python Notebook (pandas, matplotlib, SciPy, statsmodels) + Statistical Results Table; Derived Fields + Cleaned Dataset; Analytical Question Map; Dashboard UX Specification (Wireframes + Guidelines).

Exit gate: Derived fields are validated and the analytical question map is agreed.

Week 4 (1-5 November 2026): System Analysis & Design Pack and Power BI Data Models

- **Hossam:** Software Architecture, DFD and Technology Stack.
- **Yahia:** Power BI Data Models (Shipments and Hospitals, Kept Separate) and Core Measures.
- **Ola:** Problem Statement and Use Case Diagram.
- **Aya:** ER Diagram and Logical & Physical Schema.

Deliverables: Software Architecture + DFD + Technology Stack; Validated Data Models + Core Measures; Problem Statement + Use Case Diagram; ER Diagram + Logical & Physical Schema.

Exit gate: Design documents are ready for submission on 6 November 2026; relationships, denominators and filters are tested.

Week 5 (8-12 November 2026): Dashboard Build: Five Pages

- **Hossam:** Dashboard Page 2: Delivery & Cycle Time; Dashboard Page 5: Case Impact & Decision Quality.
- **Yahia:** Dashboard Page 4: Purchase Orders & Inventory.
- **Ola:** Dashboard Page 1: Executive Overview.
- **Aya:** Dashboard Page 3: Suppliers & Cost.

Deliverables: Dashboard Pages 1 to 5.

Exit gate: Every visual is linked to a specific question, column and KPI.

Week 6 (15-19 November 2026): Reconciliation, Testing, UAT and Findings

- **Hossam:** Master Reconciliation (SQL, Python, Power BI).
- **Yahia:** Test Plan, Functional Testing, UAT and Bug Log.
- **Ola:** Findings and Recommendations.
- **Aya:** README and Execution Guide.

Deliverables: Master Reconciliation Report; Test Plan + Dashboard Functional Test + UAT & Bug Log; Findings & Recommendations; README + Execution Guide.

Exit gate: No critical error remains open.

Week 7 (22-26 November 2026): Sign-off, Infographic, Presentation and Final Documentation

- **Hossam:** Technical Sign-off and Final Refresh QA.
- **Yahia:** Project Infographic (Single 16:9 Image).
- **Ola:** Project Presentation (PPT/PDF).
- **Aya:** User Manual and Final Documentation.

Deliverables: Technical Sign-off Package + Final Refresh QA; Project Infographic; Project Presentation; User Manual + Final Documentation.

Exit gate: Final internal delivery.

2.8 DEPI Submission Dates

- 16 October 2026: Project Planning & Management, Literature Review and Requirements Gathering
- 6 November 2026: System Analysis & Design
- 26 November 2026: Final internal delivery
- 30 November 2026: Implementation (Source Code & Execution)
- 4 December 2026: Final Presentation, Testing & Reports
- All documents are uploaded to GitHub.

3. Team Members and Roles

1- **Hossam - Technical Analytics Lead:** Project proposal; project plan and task assignment; KPI specification and KPIs document; GitHub repository; SQL staging and integrity checks; Python notebook (KPI recalculation, basic charts and statistical analysis); software architecture, DFD and technology stack; dashboard Pages 2 (Delivery & Cycle Time) and 5 (Case Impact & Decision Quality); master reconciliation; technical sign-off and final refresh QA.

2- **Yahia - Data Preparation & Testing Lead:** Raw and Working data layers with profiling; cleaning of the shipments and hospital tables; derived fields and cleaned dataset; Power BI data models and core measures; dashboard Page 4 (Purchase Orders & Inventory); test plan, UAT and bug log; project infographic.

3- **Ola - Data Quality & Review Lead:** Stakeholder analysis, user stories and use cases; literature review file; data quality rules and mapping sign-off; analytical question map; problem statement and use case diagram; dashboard Page 1 (Executive Overview); findings and recommendations; project presentation.

4- **Aya - Data Dictionary & Requirements Lead:** Data dictionary; functional and non-functional requirements; risk assessment; before/after reconciliation; wireframes and UI/UX guidelines; ER diagram and logical & physical schema; dashboard Page 3 (Suppliers & Cost); README and execution guide; user manual and final documentation.